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Effects of content and language integrated learning at the primary school level: A multi-level meta-analysis

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Abstract

This meta-analysis synthesized the effects of content and language integrated learning (CLIL)—an approach in which non-language subjects are taught in a foreign language—on primary school students. The dataset comprised 28 samples (N = 214,103) drawn from 21 (quasi-)experimental studies that evaluated either foreign language (FL) or academic content learning. A multi-level meta-analysis revealed that CLIL was significantly more effective than non-CLIL for FL learning (d = 0.63, SE = 0.21, p = .003), with particularly strong effects observed concerning improving speaking skills (d = 1.24, SE = 0.24, p < .001) and smaller effects for improving other language domains (d = 0.48, SE = 0.18, p = .009). Additionally, publication year was significantly associated with the impact of CLIL on FL learning, with earlier studies reporting slightly stronger effect sizes than later ones, which demonstrated marginally weaker effects (β = −0.07, SE = 0.03, p = .04). In contrast, CLIL and non-CLIL approaches were comparable in terms of aiding academic content learning (d = −0.06, SE = 0.16, p = .72). Moderator analyses suggested that the observed effects of CLIL on content learning might be influenced by pre-existing differences between groups; studies with confirmed group homogeneity indicated a negative effect of CLIL on content learning (d = −0.22, SE = 0.13, p = .09), whereas studies without confirmation of group equivalence showed a positive effect (d = 0.31, SE = 0.22, p = .17).

Keywords: Content and language integrated learning; English medium instruction; Meta-analysis; Primary school

1. Introduction

Over the past two decades, the field of language teaching and learning has witnessed increasing research on Content and Language Integrated Learning (CLIL) (e.g., Coyle et al., 2010; Eurydice, 2006; Hu et al., 2023; Li et al., 2020; Ruiz de Zarobe, 2011), an educational approach where a foreign language (FL) is used as a medium of instruction to teach non-language (i.e., content) subjects. Language educators have emphasized the value of CLIL for its authenticity (i.e., students learn non-language subjects through FL and engage in purposeful communication in authentic contexts through that language) (Coyle et al., 2010) and provide a large amount of FL input in conjunction with FL-focused instruction (e.g., traditional English-as-a-foreign-language instruction; Castellano-Risco et al., 2020; Martínez-Adrián & Gutiérrez-Mangado, 2009). Against this support, the reasons for the worldwide growth of CLIL are both proactive (e.g., promoting multilingualism in the community) and reactive (e.g., as a means of improving citizens' English proficiency) (Coyle et al., 2010), with an increasing number of countries incorporating this type of approach into their curricula (e.g., Bower et al., 2020; Gilanyi et al., 2023).

Recently, the field has received academic support to extend CLIL to primary education (e.g., Beaudin, 2022; Hood, 2020; Wang & Kirkpatrick, 2015), with a growing number of empirical studies (e.g., Fleckenstein et al., 2019; Hughes & Madrid, 2020; Jafarigohar et al., 2022; Piesche et al., 2016) aimed at measuring the effectiveness of CLIL instruction for primary school students (generally grades 1 to 6 in most educational contexts) compared with mainstream conditions, where the same academic content is taught in the students' mother tongue instead of an FL. However, these studies have yielded mixed results regarding the effects of CLIL on language and content learning, making it difficult for parents of primary school students, teachers, policymakers, and other stakeholders to assess the value of CLIL instruction. Thus, the current research on the effectiveness of CLIL in primary education calls for synthetic efforts to gauge its effect on both FL and content learning. Such efforts to gauge both types of learning simultaneously are in line with Macaro's (2018) "cost–benefit" perspective regarding this type of approach, which enables us to evaluate the possible costs associated with learning non-language subjects for their effects on FL learning.

In light of this gap in the field, the present meta-analysis was conducted to synthesize the findings of quantitative experimental studies that compared the effects of CLIL on primary school students' FL and content learning with those of mainstream conditions. Our meta-analytic efforts are in line with previous systematic reviews of the effectiveness of CLIL in both primary and secondary education (Goris et al., 2019; Graham et al., 2018) and a previous

meta-analysis of the effectiveness of CLIL for students' English learning at the secondary level (Lee et al., 2023). We also aimed to identify a set of variables that may influence the effectiveness of CLIL and examine their potential moderating effects.

We note that our meta-analysis additionally considered English Medium Instruction (EMI) studies, although the term "EMI" has been more strongly associated with Asian contexts and higher education settings, while the term "CLIL" has been mainly adopted in studies conducted in European regions. Furthermore, as Coyle et al. (2010) highlighted, "CLIL is not synonymous with English language learning and teaching" (p. 9); although English is the FL in most educational contexts owing to its status as a lingua franca, the target medium of instruction may be languages other than English (e.g., CLIL in the United Kingdom). However, it has been proposed that they share a "functionally equivalent" context (Rose et al., 2021, p. 1) with the same dual goal (i.e., teaching the content knowledge and enhancing FL learning), and thus, recent systematic reviews and meta-analysis studies have considered both CLIL and EMI studies (e.g., Gilanyi et al., 2023; Graham et al., 2018; Lee et al., 2023). Therefore, we employ the umbrella term "CLIL" to refer to both types of instruction throughout the manuscript and use "EMI" only when referring to a specific study or source. However, we did not include studies on immersion, dual-language immersion, or content-based instruction in this meta-analysis, mainly because these approaches are usually implemented in contexts where the target language is also widely used in the community (e.g., Anglophone students learning the co-official language French in Canada and Spanish-speaking students learning English in the United States). This difference in sociolinguistic context may make studies on these programs less comparable to those on CLIL and EMI.

2. Literature review

2.1. Arguments for and against CLIL in primary education

Before reviewing the empirical evidence on the effectiveness of CLIL in primary school students' learning, it is important to understand the arguments for and against starting CLIL at a young age. In the field of second language acquisition (SLA), there has been the argument that "earlier is better" (e.g., DeKeyser, 2000; Long, 1990). This argument is based on the assumption that if students begin learning a language earlier, they will be more exposed to FL. According to the input hypothesis (Krashen, 1991), more comprehensible input facilitates FL learning. This implies that if students begin CLIL at an earlier age, they will gain increased exposure to FL by learning content subjects in the FL rather than in their mother tongue, alongside regular FL lessons (e.g., Martínez Agudo, 2020; Muñoz, 2015). This, in turn, facilitates FL learning.

In addition to the sheer amount of FL input, the relationship between language and content development may favor the early start of CLIL. Johnson and Swain (1994) considered FL learning and conceptual development to be “separate but intersecting continua” (p. 212), and the theoretical model they proposed (see Fig. 1) illustrates the linguistic and conceptual development of students in different types of immersion programs. Although this model is based on immersion, its primary tenets can be transferred to CLIL contexts, as both types of programs share the key features of students learning non-language subjects through the FL they are acquiring.

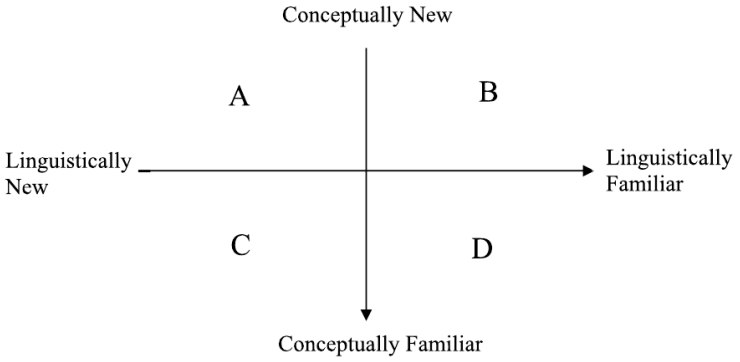


Fig. 1. Relationships between Conceptual and Linguistic Development of Immersion Students (Adapted from Johnson & Swain, 1994, p. 212).

According to this model, when students start learning content subjects in FL at an early age (e.g., 1st grade or earlier), they are in segment A (i.e., they have not acquired language or concepts). Although these students know very little about FL, they can still master the linguistic skills and knowledge of content subjects simultaneously because the language and content required for the earlier stages of education are rather simple. Therefore, these students can easily overcome the proficiency gap, which indicates the difference between their current FL proficiency levels and the threshold level required to learn the content subjects in FL.

However, when students start learning content subjects in FL at a later stage (e.g., upper grades of primary or secondary education), they may face the challenges of segment C: They have already learned concepts in their first language (L1) in primary school or their daily life but do not know how to express these concepts in FL. Therefore, they must bridge this immense proficiency gap. Thus, it seems easier for early CLIL students to master language and content simultaneously, which strengthens the argument for starting CLIL in primary education.

However, there are arguments against adopting CLIL in primary education. First, the assumption that "earlier is better" is controversial, with arguments that adults can learn languages more efficiently and strategically, and even students who start learning at an early age might not reach a native-like level of proficiency (e.g., Genesee, 2013). Hence, the effect of age on FL learning must be examined in correlation with other factors (e.g., the learning context, and learners' engagement with the language) (e.g., Muñoz, 2014). In particular, in the CLIL context, the FL input that students are exposed to is academic language, which is the language associated with representing content knowledge and is different from the everyday language that students are more familiar with (Llinares et al., 2012). It is also unclear whether and how CLIL teachers integrate content and language teaching into their lessons (Hu & Gao, 2021; Nikula et al., 2017). These factors may moderate the effects of increased FL exposure in early CLIL. Second, whether students can benefit from bilingual education programs may depend on certain conditions, for which Cummins' (1979) threshold hypothesis offers a possible explanation. This hypothesis suggests that students must achieve a certain level of proficiency in both L1 and FL to experience the cognitive and academic advantages of bilingual education (Cummins, 1979). Although the threshold level for early CLIL may be lower (Genesee, 2013; Johnson & Swain, 1994), it remains difficult to determine whether primary school students have attained the required level.

The arguments reviewed above demonstrate that there are both advantages and challenges when students begin learning content through FL in primary education. This may also explain why previous empirical studies on the effectiveness of CLIL on primary students' learning yielded conflicting results, pointing to the need for further investigation into the topic and its possible reasons.

2.2. Previous literature reviews and meta-analysis studies on the effectiveness of CLIL

In this section, we review recent literature and meta-analyses that have presented important pieces of synthetic evidence for the effectiveness of CLIL in FL and content learning. Graham et al. (2018) reviewed 25 CLIL studies, of which 10 examined primary school students' learning outcomes in terms of content and foreign language learning. Their review revealed mixed findings on the effectiveness of CLIL, with some studies showing no significant difference between the CLIL and non-CLIL groups, and others showing the superiority of CLIL. Highlighting the need for more longitudinal CLIL research, Goris et al.'s (2019) literature review included longitudinal studies examining the effects of CLIL on FL learning outcomes. The findings of the four primary education studies identified in their review did not

reveal any superiority of CLIL in FL learning. Taken together, previous literature reviews have not reached clear conclusions, which indicates that the effectiveness of CLIL in language and content learning is uncertain.

We identified two large-scale meta-analyses on the effectiveness of CLIL (Lee et al., 2023; Lo & Lo, 2014) that provided more quantitatively oriented conclusions. Lo and Lo's (2014) meta-analysis focused on the effectiveness of EMI in secondary education in Hong Kong. Their meta-analysis included 24 empirical studies that examined the effectiveness of EMI in terms of content and English language learning. Their results showed that students in EMI secondary schools outperformed those in Chinese as a medium of instruction in terms of English proficiency, while the opposite was true for academic achievement in content learning (e.g., science, history, and geography). More recently, Lee et al. (2023) conducted another meta-analysis of the effects of CLIL and EMI on secondary-level students' English learning, based on 44 samples from 38 empirical studies. Their results revealed that EMI-CLIL instruction was more effective in the development of English knowledge and skills than its mainstream counterpart, in both the short and long term. However, their results related to the moderator analyses suggest that such instruction may be less effective for some linguistic dimensions and when FL and students' L1 were not related. Lee et al. (2023) found that the superiority of CLIL instruction may be at least partially influenced by whether the primary study checked the baseline differences between the CLIL and non-CLIL groups (cf., Bruton, 2013).

Overall, the extant literature reviews and meta-analyses have yielded mixed conclusions regarding the effectiveness of CLIL instruction. Additionally, there has been little effort to use a meta-analytic approach to investigate the effectiveness of CLIL instruction for language and content learning among primary school students, a gap that the present meta-analysis aims to address. The next subsection introduces the potential moderators for the present meta-analysis and the rationale for considering them based on the aforementioned literature reviews and meta-analyses.

2.3. Potential moderator variables

2.3.1. Homogeneity Confirmation

One of the most frequently raised concerns in CLIL intervention research is the lack of confirmation of homogeneity between CLIL and non-CLIL groups (e.g., Bruton, 2013; Goris et al., 2019; Murphy et al., 2020), which could compromise the validity of the studies. In essence, at the beginning of CLIL implementation, some CLIL intervention studies may have included a CLIL group comprising students with higher FL proficiency than their non-CLIL counterparts. Although some recent studies have attempted to remedy this problem by

employing the FL proficiency test as a measure to control for baseline differences (e.g., Jafarigohar et al., 2022), including relevant confounding variables in statistical analyses (Fleckenstein et al., 2019), and counterfactual analyses (Virdia, 2022), others have not considered homogeneity confirmation in the study design or interpretation of results. Lee et al.'s (2023) recent meta-analysis of CLIL intervention studies with secondary-level students found a moderating effect of this variable, revealing that the effect size was larger when baseline differences were not controlled. Therefore, as the confirmation of homogeneity between the CLIL and non-CLIL groups has been acknowledged in the literature, we included this moderator in our meta-analysis.

2.3.2. Randomized controlled trials

Randomized controlled trials (RCTs) are a sampling method in which each potential unit has an equal chance of selection (Urdan, 2022). RCTs have been proposed as the "gold standard" in measuring the effectiveness of a particular intervention in education research (Delandshere, 2004), and as "one of the hallmarks of experimental research" (Mackey & Gass, 2005, p. 146), as it is considered the most effective method for controlling the effects of confounding variables. In CLIL experimental research, although RCTs have been deemed particularly difficult to administer, given their selective nature (Bruton, 2011), some studies have attempted to employ this sampling method at a class or school level (e.g., Martínez Agudo, 2022; Piesche et al., 2016). Therefore, this variable was used to examine whether the effectiveness of CLIL differed when RCTs were administered. This variable may, to some extent, overlap with the first moderator variable (i.e., *Homogeneity Confirmation*) in the sense that RCTs aim to control for the effects of confounding variables, including the sample heterogeneity. Nevertheless, this meta-analysis considered both moderator variables because some studies employed RCTs (Martínez Agudo, 2019, 2020; Piesche et al., 2016), making it necessary to examine both.

2.3.3. Nature of CLIL (Full CLIL curricula vs. short-term CLIL interventions)

It has been argued that the conceptualization of CLIL is "broad" and "slippery" (Cenoz et al., 2014, p. 246) because it can be implemented in various ways. For example, students in the CLIL program introduced by Agustín-Llach and Canga Alonso (2016) have been receiving the CLIL program for their science classes since 1st grade (i.e., a total of 6 years), whereas students in Piesche et al. (2016) have received only five 90-min-long lessons in CLIL. In addition to the absolute length of each CLIL program, we found that they differed in terms of their nature; the former was part of the regular curriculum, whereas the latter was a temporary

intervention as part of an experiment. These two different ways of implementing CLIL may have different impacts on students' learning and hence should be captured in the meta-analysis as a moderator variable.

2.3.4. Students' age and first language

Young primary school students undergo qualitatively different phases of linguistic, cognitive, and emotional development (Shin & Crandall, 2014). Given the characteristics of this population and the theoretical model proposed by Johnson and Swain (1994), it is expected that students in different grades of primary education will show different learning patterns related to conceptual and linguistic development when engaging in learning through CLIL instruction. Specifically, lower and upper graders may differ in terms of the linguistic and conceptual difficulty of the target learning materials, L1 literacy, FL proficiency level, and volume of schema knowledge, which influences their learning through CLIL instruction. Therefore, we considered this a moderating variable in our meta-analysis.

In addition to students' age or grade level, it is worth investigating whether the proximity of their first language (L1) to FL affects the effectiveness of CLIL instruction, because this factor may also play a significant role. In early SLA literature, this issue was addressed in contrastive analysis (CA; Odlin, 1989; Stockwell et al., 1965) research, which structurally compared and analyzed the similarities and differences between students' L1 and FL. More recently, L1–FL language relatedness has been addressed in meta-analyses of L2 reading research (e.g., Jeon & Yamashita, 2014; Lee et al., 2022) in terms of whether this moderator can influence the relationship between L2 reading comprehension and its components. Lee et al. used Beaufils and Tomin's (2020) genetic proximity calculator, according to which they coded L1-FL relatedness as either related or unrelated. In their study, L1-FL relatedness was not a significant moderator between L2 reading comprehension and its components. Nevertheless, it can be expected that the proximity between learners' L1 and FL may play some role in influencing the effectiveness of CLIL instruction and will be considered in the present study.

2.3.5. Target linguistic dimension (concerning FL learning only)

Previous review articles (e.g., Dalton-Puffer, 2008; Goris et al., 2019; Graham et al., 2018) have discussed the different effects of CLIL instruction on FL dimensions. For example, in a review of previous empirical studies, Dalton-Puffer (2008) suggested that CLIL instruction may be more beneficial for the development of receptive skills such as listening, reading, and vocabulary. Lee et al.'s (2023) meta-analysis of secondary-level students found that CLIL instruction was more effective than other learning methods for developing FL vocabulary knowledge. Among studies on primary school students, Goris et al. (2019) suggested that the

type of instructional material may place different demands on students and consequently develop different linguistic aspects. Thus, we hypothesized that the target linguistic dimension could be another important moderating variable.

2.3.6. Language for content assessment (concerning content learning only)

A recently discussed issue in the evaluation of CLIL effectiveness is the language of assessment (i.e., students' L1 or FL) for measuring content learning outcomes (Urraca, 2023). This is challenging for two reasons. First, although students' L1 appears to be a more suitable language for assessing their knowledge of an academic subject, CLIL students could be disadvantaged, as they would have learned the target concepts and notions in FL. Second, using different languages for the assessment of students under different conditions (i.e., using L1 for the non-CLIL group and FL for the CLIL group) may compromise the validity of the results obtained from such a design, as instruments designed in two different languages could not be considered the same. Existing studies vary in the language of instruments. In other words, the instruments in some studies (e.g., Fleckenstein et al., 2019; Virdia, 2022) were given to students' L1 for comprehension-related reasoning, while others designed their instruments such that the test included both languages for the CLIL group (Pérez-Cañado, 2018), or a subset of the participants received the test written in different languages. Given these variations, we examined whether the language of the instruments was a statistically significant moderator in the effectiveness of CLIL in content learning.

2.4. Research questions

Based on the literature review, the present meta-analysis addressed the following four research questions:

1. To what extent does CLIL lead to higher levels of FL learning for primary school students than its mainstream condition, which receives FL instruction?
2. How are the identified moderators related to CLIL's effect on primary school students' FL learning?
3. To what extent does CLIL lead to higher levels of content learning for primary school students than its mainstream counterpart?
4. How are the identified moderators related to CLIL's effect on primary school students' content learning?

3. Method

This section describes how a meta-analysis was conducted following the necessary steps, such as how we managed the literature search, how we constructed the dataset (i.e., data structure, effect size estimation, outlier detection, and coding scheme), and what types of data analyses were planned and employed. All the authors of the present study participated in the entire methodological journey. Any disagreements regarding dataset construction were resolved through discussion.

3.1. Literature search

3.1.1. Database and journal search

To identify previous empirical studies that focused on measuring the effects of CLIL instruction on primary school students' FL and content learning, we first conducted a database search and journal search. Regarding the database search, we visited ProQuest, Web of Science, and SCOPUS databases and searched relevant studies by typing the following combination of keywords: [EMI OR "English medium instruction" OR CLIL OR "content and language integrated learning"] AND [outcome OR development OR effect OR non-EMI OR non-CLIL OR mainstream] AND [primary OR elementary]. Specifically, the second set of keywords—"[outcome OR development OR effect OR non-EMI OR non-CLIL OR mainstream]"—aimed to identify studies that focused on measuring (a) the impact of CLIL and (b) included comparison or control groups. For part (a), terms such as "outcome," "development," and "effect" were used, while for part (b), terms such as "non-EMI," "non-CLIL," and "mainstream" were applied. The database search was filtered according to the language of the literature (written in English). In addition, to perform a manual keyword search, we selected 10 academic journals based on their impact factors and relevance to the topic. These journals included *Language Learning, Applied Linguistics, TESOL Quarterly, Language Teaching Research, System, International Journal of Bilingual Education and Bilingualism, Journal of Immersion and Content-Based Language Education, Language and Education, European Journal of Applied Linguistics,* and *International Journal of Applied Linguistics*. For these DB and journal searches, the search fields were narrowed down to the titles and abstracts of the literature. After deleting duplicates, we found a total of 337 studies at this stage.

3.1.2. Backward and forward search

Once we obtained a tentative list of studies for meta-analysis, backward and forward reference searches were performed. For the former, the reference lists of the identified empirical studies were checked to identify older empirical studies that were not included in the database search.

For the latter, we used the “Cited by” function of Google Scholar to check whether the most recent empirical studies on the tentative list were cited by younger empirical studies. After combining the results of database, journal, and reference searches, 368 studies were identified.

3.1.3. Screening based on inclusion criteria

After securing PDF files for these studies, we reviewed them to finalize the list of included studies according to the following inclusion criteria: (1) written in English, (2) focused on measuring the impact of CLIL on primary learners’ FL learning and/or content learning (i.e., learning outcome), (3) reported separate datasets for independent samples, (4) contained descriptive statistics for effect size calculations (i.e., sample sizes, means and standard deviations for Cohen’s *d*), and (5) included a comparison (or control) group in which non-CLIL instruction was given. Based on our careful review, we excluded 347 studies: (1) six were not written in English, (2) 325 did not focus on measuring the impact of CLIL on primary learners’ FL and/or content learning, (3) one included both primary- and secondary-level participants but did not report separate datasets for primary groups, (4) eight did not contain sufficient descriptive statistics for effect size calculations, and (5) seven did not include a comparison group. A total of 21 studies remained on the final list of studies included in the meta-analysis. Fig. 2 illustrates a PRIMA flowchart of the literature search.

3.2. Dataset construction

3.2.1. Understanding data structure

As the first step in constructing a dataset, we reviewed the structure of the data reported in each study and identified two important characteristics among the collected studies. First, we found that five of the 21 studies had more than one sample. For example, Anghel et al. (2016) used two independent samples, Cheng et al. (2010) used three, Jafarigohar et al. (2022) used three, Knell et al. (2007) used three, and Virdia (2022) used two. Thus, included 21 studies comprising 28 independent samples. Second, not all studies measured both language and content learning. As shown in Table 1, among the 28 samples (N = 214,103), 21 (N = 210,906) measured the effects of CLIL on students’ FL learning, nine (N = 4076) focused on measuring students’ content learning, and two (Mattheoudakis et al., 2014; Pérez-Cañado, 2018) measured learners’ FL and content learning. In terms of the dataset, it is also worth mentioning that a large portion of the current dataset consists of studies conducted in Spain (12 out of 28 samples), which should be considered when interpreting the findings of this meta-analysis (see *Limitations and Future Research Directions* section for this point).

3.2.2. Coding moderator variables

Seven potential moderator variables (i.e., homogeneity confirmation, randomized controlled trial, nature of CLIL, students' grade and first language, target linguistic dimension, and language for content assessment), along with each study's total length of instruction and publication year, were coded for further analyses of moderation effects (see Table 2 for definitions of the moderator). This meta-analysis investigates how each variable influences the overall effect of CLIL and checks whether the constructed dataset includes supporting data for moderator analyses (i.e., no missing values).

After reviewing the samples, we found that students' first language could not be included as a moderator because of limited variation; the majority (approximately 90%) were conducted in European countries, restricting insights across diverse L1 backgrounds. Therefore, we focused on six other potential variables along with the total length of instruction and publication year. The latter was not initially considered a moderator; however, we found variations across studies ranging from 2010 to 2023. As CLIL has gained increasing attention from educational communities and the importance of foreign language learning (such as English for globalized contexts) continues to grow, we believe that when a CLIL program is implemented, it may influence its impact as shifts in educational priorities and methodologies over time may shape outcomes.

As shown in Table 6, which provides a complete overview of moderator variable coding across datasets, some moderators did not have applicable categorical values in certain datasets, such as the FL learning and content learning datasets. For instance, among the 21 samples for FL learning, there were no randomized controlled trials (RCTs); therefore, the random assignment variable could not be applied to this dataset. Similarly, the 10 samples for content learning only involved upper-grade students, meaning that there were no samples for 1st, 2nd, and 3rd graders in this dataset.

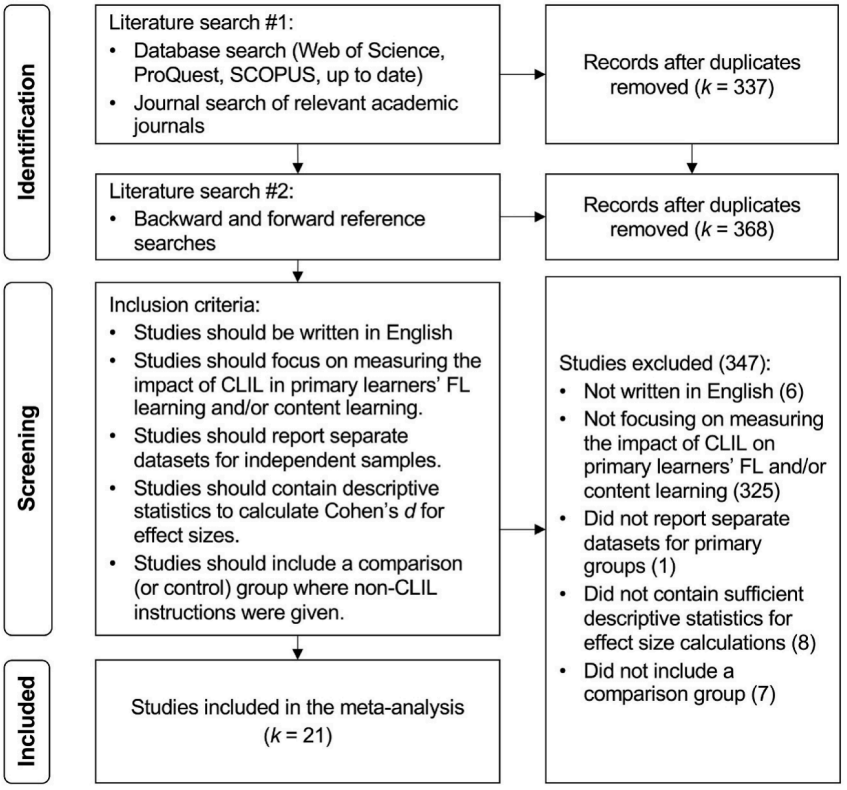


Fig. 2. Flow chart of the literature search process.

Table 1
Data structure of the collected studies.

Study	Independent Sample	Sample Size	FL Learning	Content Learning
Agustín-Llach (2017)		133	O	
Agustín-Llach & Canga Alonso (2016)		107	O	
Anghel et al. (2016)	Anghel et al. (2016) [1]	91,705	O	
	Anghel et al. (2016) [2]	96,004	O	
Canga Alonso (2015)		255	O	
Cheng et al. (2010)	Cheng et al. (2010) [1]	385	O	
	Cheng et al. (2010) [2]	430	O	
	Cheng et al. (2010) [3]	183	O	
Fernández-Pena & Gallardo-del-Puerto (2021)		63	O	
Fernández-Sanjurjo et al. (2019)		709		O
Fleckenstein et al. (2019)		590		O
Hughes & Madrid (2020)		93		O
Jafarigohar et al. (2022)	Jafarigohar et al. (2022) [1]	37	O	
	Jafarigohar et al. (2022) [2]	36	O	
	Jafarigohar et al. (2022) [3]	37	O	
Knell et al. (2007)	Knell et al. (2007) [1]	51	O	

Study	Independent Sample	Sample Size	FL Learning	Content Learning
	Knell et al. (2007) [2]	61	O	
	Knell et al. (2007) [3]	71	O	
Martínez Agudo (2019, 2020)		136	O	
Mattheoudakis et al. (2014)		51	O	O
Nieto-Moreno de Diezmas (2018)		19,860	O	
Nieto-Moreno de Diezmas & Hill (2019)		119		O
Pérez-Cañado (2018)		828	O	O
Piesche et al. (2016)		722		O
Pladevall-Ballester & Vallbona (2016)		287	O	
Uhl et al. (2023)		186	O	
Virdia (2022)	Virdia (2022) [1]	505		O
	Virdia (2022) [2]	459		O

Table 2
Coding scheme for moderator variables.

Moderator Variable	Value
Randomized Controlled Trial	(1) Randomized controlled trials (RCTs), or (2) non-RCTs, such as convenient samples
Homogeneity Confirmation	(1) Homogeneity between CLIL and non-CLIL groups was checked (e.g., statistical comparison, pre-test results, etc.) (2) Difference between CLIL and non-CLIL groups was not necessarily controlled.
Total length of instruction	How long was CLIL in years? (e.g., 1,2,3,4,5,6 years)
Nature of CLIL	(1) CLIL was introduced as part of regular curricula of the schools. (2) CLIL was introduced as an (temporary) intervention to the schools.
Students' Grade	What was students' grade? (e.g., 1st, 2nd, 3rd, 4th, 5th, 6th)
Target Linguistic Dimension	CLIL programs were conducted to enhance students' foreign language learning, particularly for one of the following dimensions: (1) Speaking, (2) Reading, (3) Listening, (4) Vocabulary, or (5) Grammar
Language for Content Assessment	(1) Foreign languages were used for assessing the impact of CLIL on content learning. (2) First languages were used for assessing the impact of CLIL on content learning.
Publication Year	The year the research article appeared in print

3.2.3. Estimating effect sizes

Estimating effect sizes involved converting the effects of CLIL programs and interventions into a standardized measure by mathematically unifying the units of analysis. After understanding the data structure of each study, the effect size of each measurement in each sample was calculated. For example, we found that Cheng et al. (2010) had two measurements, such as (1) listening and (2) reading + writing tests. Fernández-Pena and Gallardo-del-Puerto (2021) employed 10 measures to measure the different dimensions of students' speaking skills. The effect sizes and measurements are presented in Table 3. We calculated 65 effect sizes: 54 for FL learning and 11 for content learning.

3.2.4. Detecting outliers

Upon completion of the effect size calculation, we checked for outliers or influential cases among the effect sizes using *the metafor* package in R. This step was necessary to ensure the robustness and reliability of the calculated effect sizes. To this end, we used the *metafor* package in R, and this approach provides us with eight different plots, such as (1) Standardized Residuals Plot, (2) QQ-Normal Plot, (3) Cook's Distance Plot, (4) DFBETAS Plot, (5) Influence Plot (also known as Bubble Plot), (6) Covariance Ratios Plot, (7) Hat Values Plot, and (8) Weights Plot, to mathematically determine outliers and influential cases (readers are referred to the webpage of *metafor* for detailed information about these plots and mathematical backgrounds). This statistical package detects outliers by indicating cases with red dots, based on these eight criteria. Concerning influential cases that may or may not influence the overall trends, we can manually inspect the plots to determine whether a case is influential (e.g., identify cases that significantly deviate from others). If there are influential cases, we may want to conduct additional sensitivity analyses (i.e., rerunning the meta-analysis without influential cases) to determine the extent to which the removal of influential cases would change the overall results.

Table 3
Number of effect sizes and measurements for independent samples.

Samples	Number of Effect Size	Measurement
Agustín-Llach (2017)	1	Vocabulary
Agustín-Llach & Canga Alonso (2016)	1	Vocabulary
Anghel et al. (2016) [1]	1	Reading
Anghel et al. (2016) [2]	1	Reading
Canga Alonso (2015)	1	Vocabulary
Cheng et al. (2010) [1]	2	(1)Listening (2)Reading + Writing
Cheng et al. (2010) [2]	2	(1)Listening (2)Reading + Writing
Cheng et al. (2010) [3]	2	(1)Listening (2)Reading + Writing
Fernández-Pena & Gallardo-del-Puerto (2021)	10	Speaking
Fernández-Sanjurjo et al. (2019)	1	Science
Fleckenstein et al. (2019)	1	Mathematics
Hughes & Madrid (2020)	1	Science
Jafarigohar et al. (2022) [1]	2	(1)Productive vocabulary (2)Receptive vocabulary
Jafarigohar et al. (2022) [2]	2	(1) Productive vocabulary (2) Receptive vocabulary
Jafarigohar et al. (2022) [3]	2	(1) Productive vocabulary (2) Receptive vocabulary

Samples	Number of Effect Size	Measurement
Knell et al. (2007) [1]	2	(1) Speaking (2) Vocabulary
Knell et al. (2007) [2]	2	(1) Speaking (2) Vocabulary
Knell et al. (2007) [3]	2	(1) Speaking (2) Vocabulary
Martínez Agudo (2019, 2020)	4	(1) Speaking (2) Vocabulary (3) Reading (4) Listening
Mattheoudakis et al. (2014)	5	(1) Geography test #1 (2) Geography test #2 (3) Geography test #3 (4) Listening (5) Reading
Nieto-Moreno de Diezmas (2018)	6	(1) Listening subskill #1 (2) Listening subskill #2 (3) Listening subskill #3 (4) Listening subskill #4 (5) Listening subskill #5 (6) Listening subskill #6
Nieto-Moreno de Diezmas & Hill (2019)	1	Social Science
Pérez-Cañado (2018)	6	(1) Grammar (2) Vocabulary (3) Listening (4) Reading (5) Speaking (6) Natural Science
Piesche et al. (2016)	1	Science
Pladevall-Ballester & Vallbona (2016)	2	(1) Reading (2) Listening
Uhl et al. (2023)	1	Grammar
Virdia (2022) [1]	1	Science
Virdia (2022) [2]	1	Science

As shown in Fig. 3, analysis of the 21 samples that focused on students' FL learning revealed no outliers or influential cases (left plot in Fig. 3). By contrast, among the nine samples that examined students' content learning, one outlier was identified (the center plot in Fig. 3). After excluding this outlier, another red dot appears, prompting a repeated outlier removal process. This was repeated twice until no further red dots or signs of influential cases were detected (right plot in Fig. 3). Consequently, although all 21 samples for FL learning were retained, three outliers were removed from the content-learning dataset, reducing it to six samples.

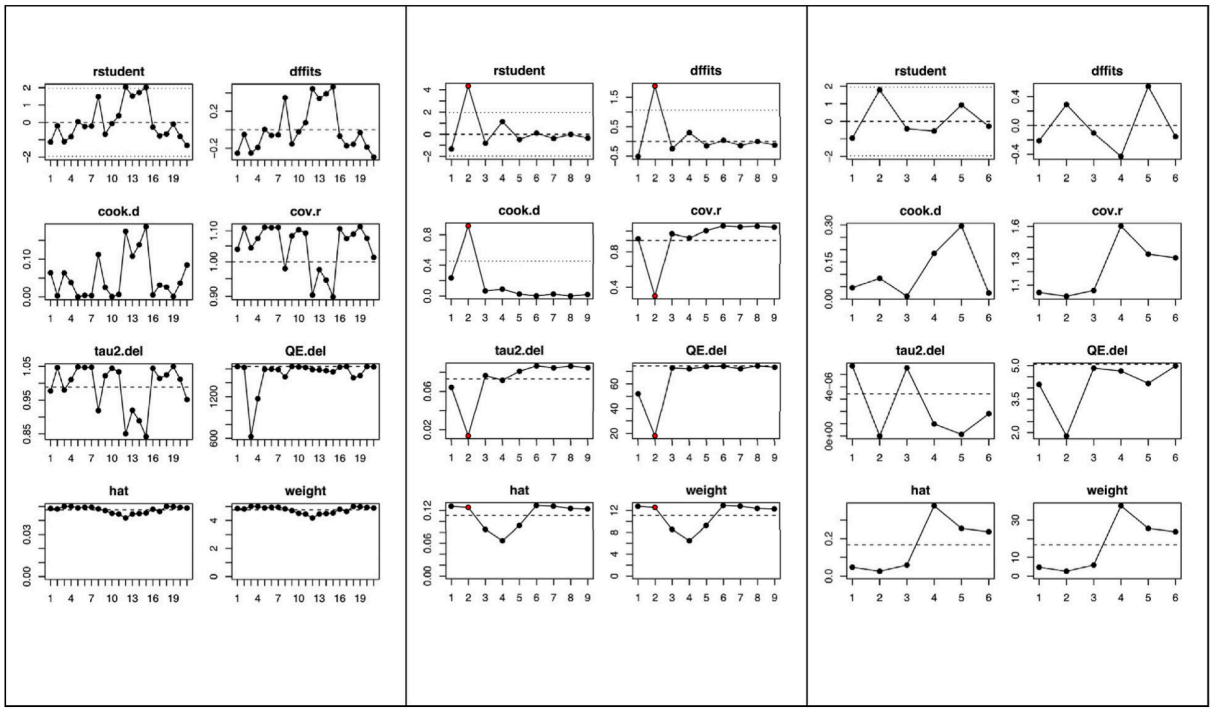


Fig. 3. Diagnosis of Outliers and Influential Cases –FL Learning (left) and Content Learning (center & right).

3.3. Dataset analysis plan

We used *Stata version 16.1* (StataCorp, 2020) to conduct further data analyses such as (1) estimating the overall mean effect size and (2) examining the moderation effects. We performed a two-level meta-analysis using the *meglm* command in Stata to build multi-level mixed-effects generalized linear models with the calculated effect sizes as their observations. This approach was adopted because we calculated the effect sizes at the measurement level, meaning that the effect sizes were nested in each sample, as mentioned in the *Estimating Effect Sizes* section. Given the dependence issue (e.g., estimating two effect sizes from one sample may contribute to a miscalculation such that this sample becomes two times more important than other samples in calculating the overall mean effect size, cf. Lee et al., 2019), calculating one effect size for each sample is required according to the traditional meta-analysis. However, as we were to use a multi-level meta-analysis, where both within-sample (Level 1) and between-sample (Level 2) variances were accounted for in the mathematical equations, independent samples can have more than one effect size while avoiding dependence. In addition, as the effect sizes are estimated values with standard errors, the variance of the effect sizes was used to replace the within-group variability of the models to account for the structures of the datasets; thus, we could assign more weight to effect sizes with larger samples (i.e., lower standard errors).

To build these models, a model with only an intercept was used to estimate the overall mean effect size, and the moderator variables were included as independent variables to examine the moderation effects. For the latter, among the eight moderator variables, the *target linguistic dimension* and *language for content assessment* were at the measurement level, whereas the remaining variables, including *students' grades, randomized controlled trial, homogeneity confirmation, total length of instruction, nature of CLIL,* and *publication year,* were at the sample level. Considering these characteristics of the current datasets, we have two equations for the data analysis, as listed in Table 4.

Table 4
Equations of multilevel meta-analysis.

Dataset	Equation
Effectiveness of CLIL on FL Learning	$Effectiveness_{ij} = \beta_0 + \beta_1(Dimensions_{ij}) + \gamma_{00} + \gamma_{01}(Random_j) + \gamma_{02}(Grade_j) + \gamma_{03}(Homogeneity_j) + \gamma_{04}(Length_j) + \gamma_{05}(Type_j) + \gamma_{06}(Year_j) + \mu_j + \epsilon_{ij}$
Effectiveness of CLIL on Content Learning	$Effectiveness_{ij} = \beta_0 + \beta_1(Assessment\ Language_{ij}) + \gamma_{00} + \gamma_{01}(Random_j) + \gamma_{02}(Grade_j) + \gamma_{03}(Homogeneity_j) + \gamma_{04}(Length_j) + \gamma_{05}(Type_j) + \gamma_{06}(Year_j) + \mu_j + \epsilon_{ij}$

In Table 4, $effectiveness_{ij}$ is the predicted effect size for the *i*-th effect size in the *j*-th sample. For Level 1, β_0 is the coefficient for the Level 1 intercept. β_1 is the coefficient for the Level 1 moderators. ϵ_{ij} is the residual term for each effect size, with its variance being the square of the estimated standard error for each effect size. For Level 2, γ_{00} is the coefficient of the Level 2 intercept. γ_{01}–γ_{06} are the coefficients of the Level 2 moderators. μ_j is the random effect at Level 2.

For our research question, the overall mean effect size can be estimated using the overall intercept ($\beta_0 + \gamma_{00}$) when all moderators at both levels are at their reference levels. Furthermore, the moderation effects can be estimated using each coefficient of the moderator variables, assuming that the other variables in the model are constant. For higher accuracy of the moderation effects, we employed additional multi-level regression models with only moderators that showed statistical significance in the first place. Therefore, the statistical significance of certain moderator variables suggests that studies with specific categorical values show a higher or lower impact of CLIL compared to those with different values within the same variable; however, this comparison is limited to each moderator variable and does not indicate which moderators have stronger or weaker effects relative to one another.

4. Results

4.1. Research question #1: Overall effects on foreign language development

Before constructing a multi-level regression model to estimate the overall mean effect size from 54 effect sizes across 21 samples, we calculated the mean effect sizes for each sample by mathematically averaging the individual effect sizes to obtain a preliminary estimation. Fig. 4 presents a forest plot of the mean effect sizes of the 21 samples, indicating that the effects of CLIL on FL learning ranged from −0.64 to 2.68. When examining the 95% confidence intervals for each sample's mean effect size, the results revealed that four samples—Uhl et al. (2023), Agustín-Llach (2017), Anghel et al. (2016) [1], and Anghel et al. (2016) [2]—had statistically significant negative effects. Additionally, six samples—Pladevall-Ballester and Vallbona (2016), Mattheoudakis et al. (2014), Fernández-Pena and Gallardo-del-Puerto (2021), Nieto-Moreno de Diezmas (2018), Martínez Agudo (2019, 2020), and Jafarigohar et al. (2022) [1]—showed null effects, as indicated by zero values within their 95% confidence intervals, whereas the remaining 11 samples reported positive effects.

To estimate the overall effects of CLIL on learners' FL learning compared to its non-CLIL counterparts, we built a multi-level regression model with 54 effect sizes using 21 samples. The results are summarized in Table 5. We found that the mean effect size of CLIL on learners' FL learning was 0.63, which was small, according to Plonsky and Oswald's (2014) benchmark suggestion for FL research (0.4, 0.7, and 1.0 for small, medium, and large, respectively). The standard error was 0.21, and the computed mean effect size was statistically significant. Therefore, we concluded that, on average, CLIL instruction had a statistically significant positive impact on improving students' FL proficiency.

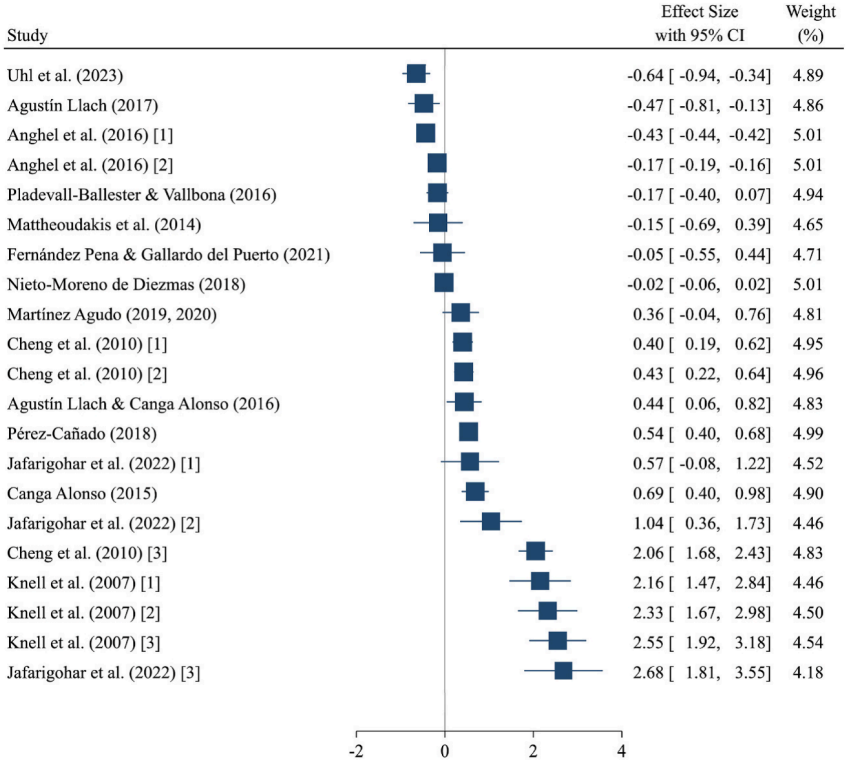


Fig. 4. Forest plot for 21 samples – FL learning.

Table 5

Results of meta-analysis for mean effect size – FL learning (*N* = 21,906).

# of ES	# of Sample	Mean ES	SE	*p*-value
54	21	0.63	(0.21)	.003

Table 6

Results of analyses for each moderator – FL learning.

Moderator Variable	# of ES	# of Sample	ES	SE	Difference
Randomized Controlled Trial					
RCT	0	0	NA	NA	NA
Non-RCT	54	21	0.63**	0.21	
Homogeneity Confirmation					
Checked	29	12	0.40	0.27	−0.55
Not checked	25	9	0.95**	0.31	
Nature of CLIL					
CLIL embedded education	49	18	0.79***	0.21	−1.10*
Short-term intervention	5	3	−0.32	0.52	
Students' Grade					
1st–3rd grades	11	6	1.15**	0.38	−0.72
4th–6th grades	43	15	0.43	0.24	
Target Linguistic Dimensions					
Speaking	15	6	1.29***	0.28	0.78*** (Speaking vs. Others)
Reading	4	4	0.55*	0.26	

Moderator Variable	# of ES	# of Sample	ES	SE	Difference
Listening	10	5	0.44	0.26	
Vocabulary	14	11	0.61*	0.24	
Grammar	3	3	0.52	0.27	

***p < .001, **p < .01, *p < .05.

Note. "Publication Year" was coded as a continuous moderator variable and was statistically significant (β = −0.09, SE = 0.04, p = .021).

4.2. Research question #2: Moderation effects on foreign language development

To understand whether our moderator variables had a statistically significant impact on the effectiveness of CLIL on primary school students' FL learning, we first built a series of regression models with each moderator as the only independent variable. As shown in Table 6, we found three moderators with statistical significance: the *Nature of CLIL, Target Linguistic Dimension,* and *Publication Year* variables. Concerning the nature of CLIL, we found an obvious distinction between CLIL-embedded education and short-term intervention (β = −1.10, SE = 0.56, p = .048). Similarly, for the *Target Linguistic Dimension,* there was a clear distinction between speaking and the other dimensions (β = 0.78, SE = 0.18, p < .001). *Publication Year,* as the only continuous moderator variable, was statistically significant, indicating a minimal tendency for earlier studies to show slightly stronger effect sizes and later studies to show slightly weaker ones (β = −0.09, SE = 0.04, p = .021). However, *the Randomized Controlled Trial* did not have sufficient values for this dataset (i.e., 21 samples of students' FL learning). *Homogeneity Confirmation* and *Students' Grade* did not show statistically significant differences across values.

For a more accurate estimation, we built a multi-level regression model with these three moderators, with statistical significance on the right-hand side of the equation. The results indicated that, on average, CLIL instructions were significantly more effective in improving the speaking dimension of FL learning than other dimensions, such as reading, listening, vocabulary, and grammar. As shown in Table 7, we found that *Publication Year* and *Target Linguistic Dimension* maintained their statistical significance in the follow-up multi-level regression (β = −0.07, SE = 0.03, p = .04; β = 0.76, SE = 0.17, p < .001, respectively), whereas the *Nature of CLIL* variable did not. Thus, we concluded that *Publication Year* and *Target Linguistic Dimension* were the only significant moderators influencing the effectiveness of CLIL in improving students' FL.

Table 7
Results of multiple meta-regression with significant moderators – FL learning.

Independent Variable	ES	SE	Difference
Nature of CLIL			
CLIL embedded education	0.77***	0.20	−0.81

Independent Variable	ES	SE	Difference
Short-term intervention	−0.04	0.46	
Target Linguistic Domains			
Speaking	1.24***	0.24	0.76***
Others (reading, listening, vocabulary, and grammar)	0.48**	0.18	

**$p < .01$, ***$p < .001$.

Note. Publication Year was added as an independent variable in the meta-regression and showed statistical significance (β = −0.07, SE = 0.03, $p = .04$).

4.3. Research question #3: Overall effects on content learning

Similar to the process used for the previous dataset, we followed the same steps for the next dataset, where the impact of CLIL instruction on learners' content learning compared with non-CLIL instruction was explored. For the six samples, we first computed the mean effect sizes for each sample to check the preliminary estimation. The forest plot in Fig. 5 indicates that the effects of CLIL on students' content learning ranged from −0.37 to 0.33. Considering the 95% confidence intervals for each sample's mean effect size, the results showed that two samples, Piesche et al. (2016) and Virdia (2022) [2], demonstrated a statistically significant negative effect, while the remaining four samples reported null effects (i.e., having zero values within their 95% confidence intervals).

To compute the overall effects of CLIL on students' content learning compared with their mainstream counterparts, we built a multi-level regression model with eight effect sizes using six samples. Table 8 presents the results of this study. The mean effect size of CLIL on students' content learning was −0.06, which is marginal. The standard error was 0.16, and the computed mean effect size was not statistically significant. Therefore, we conclude that, on average, CLIL instruction has a statistically similar impact on improving students' content learning compared to non-CLIL instruction.

4.4. Research question #4: Moderation effects on content learning

Separate regression models were developed to explore the impact of CLIL on students' acquisition of content knowledge. The overall procedure was the same as that used for RQ2. As shown in Table 9, the grade variable did not have a sufficient number of values for this dataset (i.e., 6 samples for learners' content learning).

The *Homogeneity Confirmation* moderator was the only moderator with statistical significance (β = −0.52, SE = 0.26, $p = .043$). Notable findings were observed for both categorical values. While one category had a negative overall mean ($d = −0.22$, SE = 0.13, $p = .09$), the other had a positive overall mean ($d = 0.31$, SE = 0.22, $p = .17$). That is, CLIL instruction appears to negatively impact content learning in well-controlled comparisons but not in less rigorously controlled ones. This stark contrast in outcomes from the homogeneity check moderator

highlights a significant discrepancy that could be evidence of an overestimation of the impact of CLIL instruction, as the apparent benefits might be partly due to preexisting differences rather than the instructional approach alone.

For the other moderators, we did not find any statistical significance, including *Publication Year*. Thus, we conclude that the *Homogeneity Confirmation* moderator is the only significant moderator influencing the effectiveness of CLIL in improving students' content learning. This means that homogeneity between CLIL and non-CLIL groups is a crucial factor for accurately measuring the effects of CLIL on improving students' content learning.

5. Discussion and conclusion

5.1. Overall effectiveness of CLIL instruction

The results revealed that CLIL instruction was more beneficial for primary school students in terms of FL learning than its mainstream counterpart, which consisted of only traditional FL lessons. This finding is consistent with Lee et al.'s (2023) meta-analysis of 38 empirical studies of the effects of CLIL on secondary-level students' English learning. This positive finding related to CLIL could be partly explained by the additional hours of FL exposure provided in CLIL conditions, in addition to traditional FL lessons (e.g., Agustín-Llach & Canga Alonso, 2016; Fernández-Pena & Gallardo-del-Puerto, 2021) and partly by the pedagogical contexts in which learners can engage in purposeful communication (i.e., the goal of communication is to discuss academic content rather than improve FL proficiency; Dallinger et al., 2016).

This meta-analysis also considered studies on the effectiveness of CLIL instruction for content learning, suggesting that CLIL is as effective as its non-CLIL counterparts in terms of learning academic subject content, contrary to the concerns raised regarding the use of this pedagogical approach in primary education (Halbach, 2008). However, a closer examination of the results of homogeneity confirmation as one of the moderator variables further demonstrates that CLIL instruction was indeed less effective than non-CLIL instruction in terms of content learning (see further discussion in the next section).

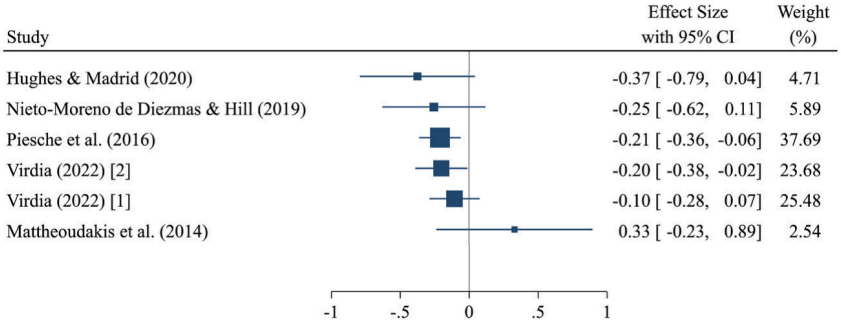


Fig. 5. Forest plot for 6 samples – content learning.

Table 8

Results of meta-analysis for mean effect size – Content learning (N = 1949).

# of ES	# of Sample	Mean ES	SE	*p*-value
8	6	−0.06	(0.16)	.72

Table 9

Results of analyses for each moderator – Content learning.

Moderator Variable	# of ES	# of Sample	ES	SE	Difference
Randomized Controlled Trial					
RCT	3	3	−0.17	0.23	−0.20
Non-RCT	5	3	−0.03	0.21	
Homogeneity Confirmation					
Checked	5	5	−0.22	0.13	−0.52*
Not checked	3	1	0.31	0.22	
Nature of CLIL					
CLIL embedded education	2	2	−0.31	0.30	0.33
Short-term intervention	6	4	0.02	0.17	
Students' Grade					
1st–3rd grades	0	0	NA	NA	NA
4th–6th grades	8	6	−0.06	0.16	
Language for Content Assessment					
CLIL language	5	3	0.03	0.20	−0.21
First language	3	3	−0.18	0.23	

*p < .05.

Note. "Publication Year" was coded as a continuous moderator variable and showed no statistical significance (β = −0.06, SE = 0.04, p = .20).

Considering the combined effects of CLIL instruction on FL and content learning, it is evident that the positive effects of CLIL on FL learning have been well documented, whereas its effects on content learning have been inconclusive. Thus, empirical studies on the effects of CLIL on primary school students to date could only partially address the "cost-benefit" question of CLIL (Macaro, 2018). However, it should be noted that studies examining the effects of CLIL on students' content learning are still scarce, probably because of the difficulty

in measuring content learning outcomes, especially in the primary education sector, where high-stakes standardized assessments are usually lacking. Hence, it is worth exploring valid research designs and measurements to address this important research question, which would inform policymakers and school leaders when deciding on an educational program or approach to be adopted.

5.2. Effects of moderator variables

In addition to the overall effects of CLIL, we examined the effects of several potential moderator variables to conduct a more fine-grained analysis of the factors affecting CLIL effectiveness. In FL learning, the target linguistic dimension was one of the significant moderator variables. In other words, CLIL was more effective in developing primary school students' speaking competence than the other linguistic dimensions (i.e., reading, listening, vocabulary, and grammar). This is different from Lee et al.'s (2023) meta-analysis on secondary learners, which revealed smaller effect sizes of CLIL on the productive aspect of English than on the overall English skills or receptive aspect of English, and larger effect sizes of CLIL on vocabulary learning outcomes. This may be explained by the natural developmental stages of FL learners, who normally develop oracy before literacy (Coelho, 2016). In the CLIL context, content subject teachers have been observed to put less emphasis on explicit language teaching (e.g., teaching grammar and different language skills) (e.g., Hu & Gao, 2021; van Kampen et al., 2018). Thus, students develop speaking skills primarily by communicating with their teachers and peers during CLIL lessons. Unexpectedly, the listening dimension did not stand out from the other dimensions as it was assumed that students were exposed to more FL input provided through the teacher's instruction. This may reflect less teacher-centered instruction or a more interactive teaching style in primary schools, where students' attention spans may be shorter than that of their counterparts in secondary schools. This may also depend on the teachers' instructional practices. If teachers make good use of non-linguistic modes such as visuals and gestures, students might not need to develop listening strategies to help them understand the instruction (Pladevall-Ballester & Vallbona, 2016).

The results for the year of publication, another significant variable, showed that more recent studies tended to report slightly smaller effect sizes for the effectiveness of CLIL for FL learning. We speculate that this trend may be due to the strengthening of FL teaching recently through various pedagogical approaches such as task-based language teaching (Long, 2015) and AI-assisted individualized learning (Lee & Lee, 2024). These advances may have reduced the relative disadvantage of learners in mainstream (non-CLIL) conditions compared with earlier years. Additionally, this finding may reflect the growing importance of English as a lingua franca driven by increasing globalization (Jenkins et al., 2018). However, we

acknowledge that our explanation lacks convincing evidence and that there may be alternative explanations for this trend. To address this issue, we call for future research to provide more detailed descriptions of the methodologies used, particularly regarding the profiles of learners in mainstream conditions and the instructions they receive. Such data could help shed light on the observed trends and provide a deeper understanding of these findings.

In terms of content learning, the variable “homogeneity confirmation” was the only significant one moderating the effects of CLIL. This finding raises two important points that are worthy of further discussion. First, it illustrates some methodological flaws in a subset of studies examining the effects of CLIL (e.g., Bruton, 2011, 2013), as the apparent benefits of CLIL might be partly due to pre-existing differences among students rather than CLIL instruction alone. This underscores the need for careful interpretation of the findings of CLIL studies based on experimental designs, especially those that compare students without confirming homogeneity before their CLIL experience. Second, assuming that included studies confirming homogeneity between two groups provide more precise and reliable results regarding the impact of CLIL on content learning, the observed negative effect of CLIL instruction on content learning (mean ES = −0.22, SE = 0.13, $p > .05$; see Table 9) aligns with Lo and Lo’s (2014) meta-analysis, which found that EMI had significant negative effects on secondary students’ learning of subjects such as science, history, and geography.

These findings further corroborate the consistently reported negative effects of CLIL on content learning. These negative effects may arise from the substantial cognitive load imposed by delivering new content-related information in a foreign language (Lorenzo, 2007). Such an approach places significant demands on students, who must simultaneously acquire both new content knowledge and foreign language skills. While Johnson and Swain (1994) hypothesized that these conceptual and linguistic challenges might be less severe for younger learners, our meta-analysis did not support this view. Instead, our findings indicate that students across educational stages face difficulties in learning academic subjects through an FL. This challenge is likely attributable to the nature of the language used in academic subjects, referred to as “academic language” or “academic literacies” (Llinares et al., 2012), which differ markedly from everyday language and pose additional obstacles for learners. In light of Cummins’ threshold hypothesis (1979), it is essential for teachers to assess students’ FL proficiency to ensure they meet a sufficient threshold for effectively acquiring academic content in an FL (Paran, 2013). However, determining such a threshold proficiency level, especially for primary learners, remains a complex issue (see for example Aizawa et al., 2023 for relevant discussion in higher education contexts). Teachers should also be mindful of the specific challenges CLIL students encounter when mastering content knowledge through FL academic language. Therefore, investigating CLIL teachers’ pedagogical strategies for

integrating content and language instruction becomes crucial. Further research should examine which teaching practices most effectively support students in learning content through an FL (e.g., Lo & Lin, 2019; Mahan, 2024).

5.3. Limitations and Future Research Directions

The present meta-analysis had several limitations. First, experimental design studies conducted in Spain accounted for a large proportion of the current meta-analysis, representing an imbalance in the dataset in terms of the target educational contexts. As previous reviews on the effectiveness of CLIL have addressed this issue (e.g., Goris et al., 2019; Graham et al., 2018; Lee et al., 2023), it is unlikely that this problem arises from the search process used in the present meta-analysis. Nevertheless, the status of the dataset in this meta-analysis calls for more experimental research in other educational contexts that should be considered when interpreting the results. Second, we separated the datasets for language and content learning, as most samples measured either content or language learning, with only two samples measuring both; therefore, the observed mean effect sizes were calculated from mostly different samples, which could be considered another limitation. Third, we were unable to include certain potential moderators in our analyses, partially because of small sample sizes and the limited information provided about these moderators in the methodology sections of the identified studies. Specifically, we suggest that future CLIL research explore moderators regarding teachers in CLIL contexts in greater depth. These moderators may include CLIL teachers' FL proficiency levels (e.g., in CEFR standards; Nieto Moreno de Diezmas & Hill, 2019), whether teachers limitedly used students' L1 (Piesche et al., 2016), and the availability of general teachers in CLIL lessons (Fernández-Sanjurjo et al., 2019), to list just a few. While our analysis revealed the moderating effect of "homogeneity confirmation," we acknowledge that most studies have tended to confirm homogeneity in students' TL proficiency instead of their content knowledge. This also serves as an important moderator of the effect of CLIL on student learning. Investigating these moderators is expected to enable us to gauge the effectiveness of CLIL more accurately and provide more detailed pedagogical implications.

Despite these limitations, as one of the first studies to quantitatively synthesize the findings of existing CLIL studies with primary school students, the results of the present study are expected to make a significant contribution to the literature on CLIL education in terms of both theoretical and practical implications and to advance our understanding of the CLIL effectiveness.

Originality and exclusivity

This manuscript is original, has not been previously published, and is not under consideration for publication elsewhere.

Author statement

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Authors: Jang Ho Lee, Hansol Lee, Yuen Yi Lo.

Authorship contributions

o Jang Ho Lee: Conceptualization, Supervision, Writing – Original Draft, Review & Editing

o Hansol Lee: Methodology, Data Analysis, Writing – Original Draft, Review & Editing

o Yuen Yi Lo: Writing – Original Draft, Review & Editing.

Ethical compliance

The research was conducted in compliance with ethical guidelines.

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Conflict of interest

The authors declare no conflicts of interest related to this manuscript.

Data availability

Data will be made available on request.

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References marked with an asterisk indicate studies included in the meta-analysis.

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